PROJECT OVERVIEW

KIIT Transit

Live tracking and trip planning for KIIT buses, EV shuttles and the Mo Bus city network.

STATUS
V1 built, in testing

PLATFORM
Flutter (Android, web)

UPDATED
23 October 2026

01 What KIIT Transit is

KIIT Transit is a mobile app that shows students where every KIIT bus and electric shuttle is right now, how long it will take to reach them, and how to get from one campus, or one part of the city, to another.

KIIT University spreads its 20 to 25 campuses across about 25 square kilometres of Bhubaneswar. Students move between them all day for classes, labs, events and hostels, but the bus and shuttle system has no published timetable, no live information and no single map. Vehicles often leave only when their seats fill, which can mean a 2-minute or a 30-minute wait. New students in particular have no reliable way to know which vehicle goes where.

The app has two sides that share one live data feed:

- **The student app** shows the live map, nearest vehicle and its arrival time, routes, bus schedules, a trip planner and a way to signal that a group is waiting for a ride.
- **The driver app** lets a driver pick the vehicle they are operating, start a duty and broadcast its GPS position every four seconds, even with the screen off. It also shows where students are waiting.

02 The final goal

The end state is a dependable, campus-wide transport companion that students open instead of guessing, and that the transport office can use to see demand. In concrete terms:

Goal	What it means in practice
Live visibility	Every on-duty KIIT bus and EV shuttle appears on the map within seconds, with an arrival estimate from where the student is standing.
One trip planner	Any two points, campus to campus or across Bhubaneswar, Cuttack, Khordha and Puri, give direct options first, then trips with one or two changes, using KIIT vehicles and Mo Bus together.
Demand signals	Students waiting in a crowd can tell nearby drivers, so vehicles go where they are needed rather than waiting to fill up.
Production backend	Real accounts limited to @kiit.ac.in emails, driver roles managed by the admin, data stored and streamed through Supabase.
Later phases	An admin dashboard for the transport office, push notifications, and Google sign-in.

03 What we are using

The app is a single Flutter codebase. The table lists every main tool and why it was chosen.

Area	Tool	Role in the project
App framework	Flutter 3.38 (Dart)	One codebase for Android, the web build and later iOS.
Maps	flutter_map 8 + MapTiler tiles	Open-source map widget on OpenStreetMap data; MapTiler serves the tiles.
Backend	Supabase (Postgres, Auth, Realtime)	User accounts, vehicles, routes, schedules and demand requests, with row-level security and live updates.
Location	geolocator + Android foreground service	Driver GPS every 4 seconds, kept alive when the phone is locked.
App structure	provider, go_router	MVVM state management and role-based navigation (students and drivers see different apps).
Design	Google Stitch, "Campus Pulse"	Sage green palette, Plus Jakarta Sans and Inter fonts, rounded card layout.
Network data	Python build script	Turns hand-transcribed routes into map data: geocoding with MapTiler and Nominatim, road snapping with OSRM.
Testing	flutter_test, headless Chrome	21 automated tests; screens checked in a browser build.

How the code is organised

The app follows a layered MVVM pattern. Screens (views) read from view models; view models call repositories; repositories talk to a single data source. That data source is either the real Supabase backend or a built-in demo backend with simulated vehicles, chosen automatically at start-up. This is why the app can be explored without any server: if no Supabase keys are supplied, it runs in demo mode.

> **Demo mode**
> Any password of six or more characters works. `driver@kiit.ac.in` opens the driver app, `admin@kiit.ac.in` is an admin, and any other @kiit.ac.in address is a student.

04 Where the data comes from

Three sources feed the app. Each has limits that shape what the app can honestly show.

Source	What it gives us	Limits
Driver phones (live)	Position, heading and speed of each on-duty vehicle.	Only exists once drivers use the app on real phones.
KIIT field research (codedex.io)	22 campuses and 38 two-way bus and shuttle links, recorded over 12 days of travelling between campuses.	No timetables: vehicles leave when full, after 2 to 30+ minutes.
CRUT Mo Bus network map (PDF)	27 city bus routes (10 to 70, including 29E) and their stop order across Bhubaneswar, Cuttack, Khordha, Puri and Konark.	A schematic with no coordinates and no departure times.

The Mo Bus routes were transcribed stop by stop from the map image. Stop positions were then looked up with two geocoding services and checked against neighbouring stops. About 120 of 383 Mo Bus stops have a confirmed position; the rest are estimated along the route between them and are marked as approximate inside the app. Only Campus 3, 6 and 15 have exact positions in public map data; the other campus positions were placed by hand.

05 What has been done

Work so far has happened in three stages.

Date	Stage	Outcome
28 Sep 2026	Version 1	Student and driver apps, authentication, Supabase schema and demo backend, built from the two specification PDFs.
2 Oct 2026	Design match	Whole app restyled to the Google Stitch design; new Home dashboard and Routes tab.
10 Oct 2026	City network	KIIT campus links and the Mo Bus network added, traced on the map, with a trip planner across both.

Student app

- **Home:** greeting, nearest vehicle with a large arrival estimate, shortcuts, a "Plan a trip" entry and a live activity feed.
- **Live map:** animated bus and EV markers with heading arrows, filters by vehicle type, distance and arrival time for each vehicle, upcoming stops.
- **Routes:** three lists (KIIT links, Mo Bus, live campus loops). Each route opens a map with its line traced along real roads and its full stop list.

- **Trip planner:** choose From, To and an optional stop on the way. Direct rides come first, then trips with one or two changes, including short walks between nearby stops. Each leg shows what to expect at the boarding stop.
- **Schedules:** next departure for timetabled campus buses, marked on time, delayed or cancelled.
- **Request a ride:** signal that students are waiting at a location; requests from the same place are grouped and expire after 30 minutes.
- **Driver directory:** drivers, their vehicles, duty status and a call button.

Driver app

- Vehicle selection, start and end duty, and a live tracking screen with speed, accuracy and update status.
- GPS broadcast every four seconds through an Android foreground service, so tracking continues with the screen off.
- A map and list of nearby student demand, with the option to mark a request as served.

Platform and quality

- Sign-in limited to @kiit.ac.in addresses, with role-based routing for students, drivers and admins.
- Supabase schema written: tables, row-level security rules, realtime feeds and the request-merging function, with seed data.
- 21 automated tests pass and the code analyser reports no issues. The web build has been checked screen by screen in a browser.

> **Not done yet**
> The Supabase backend has not been run against a real project, the app has not been tested on physical phones, and no Android installer (APK) has been built. These, along with checking map positions, are covered step by step in the companion document, "Roadmap and How-To Guide".